

Customer Segmentation for an Italian Fashion Retailer

A Data-Driven Approach Using UMAP and Gaussian Mixture Models

Group Project Report

Data Science in Action — LUISS Guido Carli

In partnership with Jakala S.p.A. — JScience Advanced Analytics

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Business Case: Customer Segmentation · Fashion Retailer Company

Dataset: 21,477 customers · March 2023 – February 2025

Revenue (observed): €11.96M · **CLV (projected):** €15.3M

Methodology: RobustScaler → UMAP (3D) → GMM (K=6)

This report presents the complete methodology, technical decisions, statistical evaluation, and business implications of a customer segmentation project developed for a large Italian fashion retailer. It is submitted as the final group assessment deliverable for the Data Science in Action module at LUISS Guido Carli, in the context of the Jakala business case challenge. Every analytical and methodological decision documented herein is justified with reference to both technical criteria and business requirements.

Abstract

This report documents a complete customer segmentation project conducted on the transactional database of an anonymised Italian fashion retailer. The dataset encompasses 21,477 registered customers, approximately 24 months of transactional history (March 2023 – February 2025), and four source tables: customer anagraphics, line-level transactions (including returns), product catalogue, and email marketing interactions. Following the CRISP-DM framework, we engineered 32 behavioural features across six dimensions — Recency/Frequency/Monetary (RFM), discount propensity, email engagement, return behaviour, category affinity, and channel mix — before applying RobustScaler normalisation and UMAP dimensionality reduction to three components. A Gaussian Mixture Model with K=6 components, selected via Bayesian Information Criterion (BIC) optimisation and validated against operational constraints, identified six behaviourally distinct customer segments. Statistical evaluation using Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Score confirms cluster validity. The six segments — The Inner Circle, Rising Stars, Style Explorers, Deal Chasers, Dormant Potential, and Quality Seekers — exhibit substantially different behavioural profiles that prescribe distinct marketing levers, enabling the transition from mass marketing to targeted, data-driven customer engagement. A financial modelling framework projects €15.3M in discounted Customer Lifetime Value, with a proposed €140K/year strategic investment expected to yield +€718K/year in incremental revenue, corresponding to a 5.1× gross return multiple.

Keywords: customer segmentation, unsupervised learning, Gaussian Mixture Model, UMAP, RFM, Customer Lifetime Value, fashion retail, CRISP-DM, BIC

1. Introduction

1.1 Digitalisation and the Strategic Value of Customer Data

The digitalisation of the retail sector has fundamentally altered the relationship between firms and their customers. Physical point-of-sale interactions, once the primary source of purchase data, have been supplemented — and in many cases supplanted — by digital touchpoints that generate high-frequency, granular behavioural signals: transaction timestamps, product-level purchase records, promotional exposure and response, returns, and email marketing interactions. The aggregate of these signals constitutes a strategic intangible asset whose value lies not in its raw form, but in the firm's capacity to extract actionable intelligence from it.

Customer segmentation — the partitioning of a heterogeneous customer base into homogeneous groups sharing similar behavioural profiles — represents one of the most direct mechanisms by which firms convert data assets into competitive advantage. Modern machine learning approaches allow for the discovery of naturally occurring clusters in high-dimensional behavioural spaces without imposing pre-defined thresholds, a shift from rule-based to data-driven segmentation that exemplifies the broader productivity gain associated with AI adoption in marketing operations. Crucially, this productivity gain is not simply about generating insights faster — it is about **allocative efficiency**: redirecting budget from undifferentiated mass campaigns toward interventions calibrated

to each segment's specific behavioural profile and economic potential. A uniform marketing strategy applied to a structurally heterogeneous customer base is, by definition, an allocatively inefficient use of resources — it simultaneously over-serves low-value segments and under-invests in the highest-value cohorts.

1.2 Business Problem

The client is a large Italian fashion retailer selling clothing and accessories. Despite operating a CRM programme that has generated 24 months of transactional data covering 21,477 customers and €11.96M in observed revenue, the retailer has historically employed a uniform, one-size-fits-all marketing approach. This ignores the structural heterogeneity of the customer base: high-value customers, discount-dependent buyers, occasional purchasers, and dormant accounts receive identical communications and promotional treatment, leading to suboptimal retention, margin erosion, and misallocation of marketing resources.

1.3 Research Objectives

- **RQ1:** Can unsupervised machine learning identify behaviourally distinct, interpretable, and actionable customer segments from the available transactional data?
- **RQ2:** What are the key behavioural characteristics that differentiate each segment, and what marketing strategies do they prescribe?
- **RQ3:** What is the estimated financial impact of deploying segment-specific strategies relative to the current undifferentiated approach?

1.4 Report Structure

The report follows the CRISP-DM framework: Section 2 (Theoretical Background), Section 3 (Data Understanding), Section 4 (Data Preparation and Feature Engineering), Section 5 (Dimensionality Reduction), Section 6 (Clustering Methodology), Section 7 (Model Evaluation), Section 8 (Segment Profiles), Section 9 (Business Implications), Section 10 (Deployment), Section 11 (Limitations), Section 12 (Conclusions).

2. Theoretical Background

2.1 Customer Segmentation

Customer segmentation is a foundational concept in marketing science, rooted in the recognition that a firm's customer base is intrinsically heterogeneous. The objective is to partition customers into mutually exclusive groups that are internally homogeneous — sharing similar needs, behaviours, or value profiles — while being maximally distinct from one another. The managerial rationale is that distinct segments respond differently to marketing stimuli, and tailoring communication, pricing, and product recommendations to each segment yields superior outcomes relative to undifferentiated mass marketing.

2.2 The RFM Framework

Recency, Frequency, and Monetary (RFM) analysis, formalised by Bult and Wansbeek (1995) and widely operationalised in CRM practice, provides a compact behavioural summary of customer purchase patterns. **Recency** (days since last purchase) captures engagement currency and churn risk. **Frequency** (number of distinct purchase occasions) reflects loyalty and habit formation. **Monetary** value (total spend or average order value) quantifies direct economic contribution. Together, these three dimensions account for a large proportion of variance in customer value and form the core of the feature matrix in this project, supplemented by five additional behavioural dimensions.

2.3 Customer Lifetime Value

Customer Lifetime Value (CLV) projects the expected discounted future revenue a customer will generate over their relationship with the firm. The discounted CLV formula employed in this project is:

$$dCLV = AOV \times freq_annual \times (retention_rate / (1 + r - retention_rate))$$

where $r = 10\%$ serves as a WACC (Weighted Average Cost of Capital) proxy. This formulation is consistent with the Pareto/NBD and BG/NBD literature (Fader et al., 2005) in treating CLV as a function of frequency, monetary value, and retention probability, while remaining computationally tractable at scale. The WACC proxy of 10% reflects a conservative discount rate appropriate for a medium-sized Italian retailer operating in the current macroeconomic environment.

2.4 Unsupervised Learning for Customer Analytics

Unsupervised learning methods discover latent structure in feature space without labelled training data. In the customer analytics context, they are preferred over supervised classification because (i) ground-truth segment labels do not exist a priori, and (ii) the objective is discovery rather than prediction. The choice among algorithms involves trade-offs between computational efficiency, cluster shape assumptions, robustness to outliers, and interpretability — all of which are examined in Section 6.

2.5 CRISP-DM Framework

The Cross-Industry Standard Process for Data Mining (CRISP-DM) provides a structured, iterative framework for data science projects, comprising six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment. This report follows the CRISP-DM structure both as an organisational device and as a quality standard: each phase is

explicitly documented, decisions are justified with reference to both technical criteria and business requirements, and the Deployment phase is addressed in Section 10.

3. Data Understanding

The retailer provided four pipe-separated text files covering the period March 2023 to February 2025. All four sources share a common customer identifier (CRM ID), enabling unambiguous record linkage across tables.

3.1 Source Tables

Source	Granularity	Key Fields	Role in Project
Anagraphics	1 row / customer	gender, date_of_birth, province, subscription_date	Demographic descriptors: age, geographic index, customer tenure
Transactions	N rows / customer (includes returns)	date, store_id, product_id, quantity, list_price, discount_pct, amount_spent	Core behavioural features: RFM, discount propensity, return behaviour, channel
Products	1 row / product	product_id, category, subcategory, gender	Category affinity features: n_categories_explored, top_category_share
Marketing Campaigns	N rows / customer	date, email_id, opened (Y/N), clicked (Y/N)	Email engagement features: open_rate, click_rate, engagement_score

Table 1: Summary of the four source datasets provided by the client.

3.2 Scope and Coverage

The merged dataset covers **21,477 unique customers** with at least one recorded purchase within the observation window. The transaction log explicitly encodes returns as negative-quantity line items, providing behavioural data on reverse logistics. Email campaign data covers only newsletter subscribers; non-subscribers are treated as zero-engagement rather than missing.

3.3 Data Quality Assessment

A systematic audit was performed prior to feature engineering. Four material data quality issues were identified:

- **Negative net spend:** A subset of customers had `return_value > gross_spend`, yielding economically impossible negative net monetary values. These customers were flagged for removal (resolution in §4.1).
- **Missing recency:** Customers with no recorded transactions in the window were assigned `recency_days` equal to the dataset maximum, representing maximum dormancy rather than absence.
- **Newsletter non-subscribers:** All email engagement features assigned 0, consistent with treating non-subscription as zero digital engagement rather than as missing data.
- **Extreme outliers:** A small number of customers with anomalously high monetary values (likely corporate accounts or data entry errors) required winsorization prior to scaling (§4.5).

4. Data Preparation and Feature Engineering

Feature engineering is the most consequential step in this project. The choice of input features determines which dimensions of customer behaviour the model can distinguish. The guiding principle was that features should capture independent dimensions of behavioural variation — not correlated proxies of the same construct — and each feature must be interpretable in a business context.

4.1 Removal of Negative Spend Anomalies

Customers whose total `return_value` exceeds `gross_spend` exhibit a negative `net_spend`. This is economically impossible under standard retail operations and signals either data entry errors or returns processed against purchases outside the 24-month observation window.

Decision: These customers were removed from the feature matrix. **Justification:** Retaining them would introduce a spurious 'negative monetary' cluster that reflects data inconsistency, not genuine customer behaviour. A clustering model trained on corrupted records would produce unreliable segment assignments for all customers, not just the anomalous ones, because the contaminated records shift the position of cluster centroids in feature space. The correct resolution is removal at the data preparation stage, with a recommendation to the client to investigate the root cause in the source transaction system.

4.2 One-Timer Treatment

One-timers are customers with exactly one purchase occasion in the 24-month window. They represent a legitimate behavioural segment: they have demonstrated purchase intent but have not developed repeat purchase behaviour.

Decision: One-timers were **retained** and flagged with `is_one_timer = True`. **Justification:** Removing them would bias the clustering toward active customers and systematically prevent the identification of the Dormant Potential segment — a commercially important group for low-cost reactivation campaigns. The `is_one_timer` flag is classified as a post-hoc descriptive feature (used to characterise clusters after formation) rather than a clustering input, avoiding the distortion that a binary flag would introduce into the continuous behavioural feature space.

4.3 The Clustering-vs-Description Distinction

A critical architectural decision in the feature engineering pipeline is the separation of features into two categories: (i) **clustering features** — continuous behavioural metrics used as direct inputs to the scaling and dimensionality reduction pipeline — and (ii) **descriptive features** — variables used to characterise and interpret clusters post-hoc but excluded from the clustering input. Binary flags (`is_one_timer`, `has_newsletter`), raw counts (`nl_count`, `n_items_bought`), and financial aggregates already captured by derived features (`gross_spend`, `net_spend`) fall into the descriptive category. Mixing binary and continuous features in a Euclidean distance metric creates dimensional dominance: a binary flag whose variance is bounded by 0.25 will be overwhelmed by continuous features with standard deviations of several hundred euros, even after scaling. The `_CLUSTERING_EXCLUDE` list in `data_processing.py` enforces this separation systematically.

4.4 The 32-Feature Matrix: Six Behavioural Dimensions

Dimension	Key Features	Business Rationale
RFM	recency_days, frequency, monetary, aov, freq_annual	Core purchase intensity and value. Recency captures engagement currency and ch
Discount Propensity	avg_discount_pct, pct_items_discounted, discount_weighted_spend	Measures price sensitivity and promotional dependency. High values identify custom
Email Engagement	open_rate, click_rate, engagement_score	Captures responsiveness to digital CRM communication. Distinguishes actively eng
Return Behaviour	return_rate, return_value_ratio, has_returns	High return rates signal information asymmetry (fit/quality uncertainty) or opportunis
Category Affinity	n_categories_explored, top_category_share, cross_category_ratio	Measures breadth of product interest. High breadth signals cross-category affinity a
Channel Mix	n_stores_visited, ecommerce_share	Captures omnichannel engagement. Distinguishes digitally native customers from i

Table 2: Feature engineering — six behavioural dimensions and business rationale.

4.5 Outlier Handling: Winsorization at the 99th Percentile

Winsorization caps values above a specified quantile at that quantile's value, preserving all records while reducing the disproportionate influence of extremes. It is preferred over trimming (which removes records) because every customer in the database represents a real individual whose segment assignment matters for campaign targeting.

Why the 99th percentile specifically: The 95th percentile would aggressively cap the top 5% of customers, including the genuine high-value core of The Inner Circle segment — the very group whose distinct profile we need to preserve. The 100th percentile offers no protection against data errors. The 99th percentile surgically removes genuine anomalies (corporate accounts, data entry errors above realistic purchase thresholds) while preserving the full behavioural range of the top 1% of legitimate high-value customers. Applied to: monetary, average order value, and return_value.

4.6 Feature Scaling: RobustScaler

Clustering algorithms operating on Euclidean distances — including UMAP's graph construction and GMM's covariance estimation — are sensitive to the relative scale of input features. Without scaling, features measured in euros (monetary, ranging from tens to thousands) would dominate over normalised ratios (return_rate, ranging from 0 to 1).

Why RobustScaler over alternatives: StandardScaler (mean $\pm$ standard deviation) is sensitive to residual outliers that survive winsorization, because extreme values in the tail inflate the standard deviation used for normalisation. MinMaxScaler maps features to [0, 1] but is equally sensitive to any surviving extremes. RobustScaler uses the median and interquartile range (IQR) for centring and scaling, making it inherently robust to non-Gaussian, right-skewed distributions of the kind characteristic of customer spend data. The theoretical justification is that for heavy-tailed distributions, the median is a more robust location estimator than the mean, and the IQR is a more robust scale estimator than the standard deviation.

5. Dimensionality Reduction: UMAP

5.1 Motivation

With 32 input features after engineering, the feature matrix occupies a high-dimensional space in which Euclidean distances lose discriminative power — the curse of dimensionality (Bellman, 1961). As dimensionality increases, the ratio of maximum to minimum pairwise distance approaches 1, making it progressively difficult for distance-based algorithms to identify meaningful cluster structure. Additionally, several of the 32 features exhibit linear or non-linear correlations: frequency and monetary value are positively correlated for high-value customers; avg_discount_pct and pct_items_discounted measure overlapping constructs. Dimensionality reduction compresses this redundancy while preserving the independent variance that drives segment separation.

5.2 Algorithm Comparison

Algorithm	Mechanism	Limitation	Verdict
PCA	Linear projection maximising variance along orthogonal axes	Cannot capture non-linear behavioural manifolds. Testing confirmed >32% of variance in first 2 components but no meaningful cluster separation in the resulting 2D space.	Rejected
t-SNE	Non-linear; optimises local neighbourhood preservation in 2D	No transform() method — cannot score new customers without rerunning the full algorithm. Does not preserve global structure. Sensitive to perplexity hyperparameter.	Rejected (visualisation only)
UMAP	Non-linear; Riemannian geometry and algebraic topology; preserves both local and global structure	Minor stochastic variation across hardware environments even with fixed random_state (addressed via seed and documentation).	Selected

Table 3: Dimensionality reduction algorithm comparison.

5.3 UMAP Configuration and Rationale

Parameter	Value	Justification
n_components	3 (clustering) 2 (visualisation)	Three components retain substantially more of the original manifold structure than two, improving clustering performance.
n_neighbors	30	Controls the trade-off between local and global structure preservation. Low values (5–15) overfit to local noise; high values (>50) lose local structure. 30 is a robust default.
min_dist	0.1	Minimum distance between points in the embedded space. The standard recommended value for UMAP is 0.1.
metric	euclidean	Standard metric for scaled continuous features. Manhattan and cosine metrics were tested and found inferior for this dataset.
random_state	42	Ensures reproducibility of the spectral initialisation. Without a fixed seed, different runs may produce different embeddings.

Table 4: UMAP hyperparameter configuration and justification.

5.4 The Hairball Problem — Why Density-Based Methods Failed

Prior to selecting UMAP as the dimensionality reduction framework, DBSCAN and OPTICS were evaluated directly on the scaled 32-feature matrix. Both algorithms require that clusters be separable by regions of low point density — density gaps. Customer behavioural data, however, exhibits near-uniform density in feature space: there are no sharp boundaries between customer types, only gradual transitions along continuous behavioural dimensions. A customer on the boundary between The Inner Circle and Rising Stars is genuinely an intermediate case, not an outlier to be classified as noise.

In practice, for every epsilon value tested with DBSCAN across three orders of magnitude, over 99.7% of customers collapsed into a single mega-cluster, with only a handful of extreme outliers assigned to separate components. This empirical result — informally termed the 'hairball problem' in reference to the undifferentiated mass of points it produces in visualisation — is not a failure of DBSCAN per se, but rather a reflection of the data's intrinsic topological structure. It directly motivated the selection of GMM, a probabilistic model that does not require density gaps and instead fits overlapping Gaussian distributions to a continuous data manifold.

6. Clustering Methodology

6.1 Algorithm Selection

Algorithm	Core Assumption	Why Rejected / Selected
K-Means	Spherical clusters of equal variance; hard assignment	Customer segments are not spherical. A high-spending, high-frequency customer may be
Hierarchical Clustering	Hierarchical tree structure; agglomerative bottom-up	$O(n^2 \log n)$ time complexity is computationally prohibitive at 21,477 records. The dendrogr
DBSCAN	Density-connected clusters; separated by low-density regions	Core ball problem (§5.4): customer behavioural data has no density gaps. All epsilon values
OPTICS	Variable-density extension of DBSCAN	Same fundamental density-gap requirement as DBSCAN. Same failure mode in the absence
GMM	Data drawn from a mixture of multivariate Gaussian distributions; soft assignment	Flexible covariance structure allows ellipsoidal, overlapping cluster geometries. Soft assign

Table 5: Clustering algorithm comparison and selection rationale.

6.2 GMM Configuration

A Gaussian Mixture Model treats the data as a superposition of K multivariate Gaussian distributions, each characterised by a mean vector, covariance matrix, and mixing weight. The Expectation-Maximisation (EM) algorithm iteratively computes soft responsibilities (probabilities of cluster membership) and updates component parameters until convergence.

Parameter	Value	Justification
covariance_type	'full'	Each GMM component has its own unconstrained covariance matrix, allowing clusters to have
n_init	10	The EM algorithm is susceptible to convergence at local optima depending on random param
random_state	42	Fixed seed for reproducibility across environments.
n_components	6	Determined by BIC optimisation and business validation (§6.3).

Table 6: GMM configuration and justification.

6.3 K Selection via BIC Optimisation

The number of GMM components K is the primary hyperparameter requiring external specification. Two complementary criteria were used:

- Statistical criterion — Bayesian Information Criterion (BIC):** $BIC = k \cdot \ln(n) - 2 \cdot \ln(L)$, where k is the number of free parameters (proportional to $K^2 \times d$ for `covariance_type='full'`) and L is the maximised likelihood. BIC penalises model complexity more strongly than AIC, making it the preferred criterion when parsimony and interpretability are priorities. The BIC curve was computed for $K = 3$ to $K = 10$. The elbow — the point beyond which additional components yield diminishing BIC improvements — was identified at $K = 6$. The AIC curve confirmed a consistent pattern, providing convergent statistical evidence.

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- **Operational constraint:** The marketing team can realistically develop, execute, and monitor at most 6 simultaneous segment-specific campaigns within a given budget cycle. $K > 6$ produces segments that are statistically distinguishable but not operationally distinct — the incremental difference between $K=7$ and $K=6$ segments does not translate into a materially different marketing lever, budget allocation, or campaign brief.
 - **Interpretability validation:** At $K=6$, all six segments exhibit clearly differentiated radar chart profiles, and no two segments share an identical optimal marketing strategy. Testing $K=4$ confirmed the loss of the Inner Circle / Rising Stars distinction: these two segments merged into a generic 'active customers' group, eliminating the most commercially critical differentiator (VIP retention vs. upgrade path activation).

6.4 Cluster Consolidation: 10 Components → 6 Segments

An exploratory GMM with 10 components was run to examine raw cluster structure before applying the $K=6$ constraint. Four components (2, 3, 6, 8) exhibited near-identical behavioural profiles: low purchase frequency, high recency (extended inactivity), and low average order value. Pairwise Euclidean distances in UMAP coordinates confirmed their proximity. These four were consolidated into the single *Dormant Potential* segment. Two mid-tier components (0 and 9) showed near-identical KPIs across all six behavioural dimensions and were merged into *Rising Stars*. The $K=6$ final model directly encodes this consolidated structure, with BIC confirming that the merged model is statistically preferable to the 10-component version.

7. Model Evaluation

The K=6 GMM was evaluated using three standard internal clustering validation metrics. Internal metrics assess cluster quality using only the data, without external labels, making them appropriate for unsupervised segmentation.

Metric	Formula / Principle	Interpretation
Silhouette Score	$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ a(i) = mean intra-cluster distance b(i) = mean nearest-cluster distance	Range [-1, 1]. Higher = better separation and cohesion. Values > 0.5 indicate good clustering.
Davies-Bouldin Index (DBI)	Average ratio of intra-cluster scatter to inter-cluster separation for each cluster pair	Lower = better. DBI = 0 indicates perfect separation. Values < 1.0 indicate good clustering.
Calinski-Harabasz Score (CHI)	Ratio of between-cluster variance to within-cluster variance, weighted by cluster sizes	Higher = better; no upper bound. A higher CHI indicates compact clusters (low within-cluster variance) and well-separated clusters (high between-cluster variance).

Table 7: Clustering evaluation metrics. Values computed in final_segmentation.ipynb.

Important interpretive note: Silhouette scores above 0.5 are typically achievable only when cluster boundaries are sharp and well-defined. Customer behavioural data is characterised by gradual, continuous transitions: a customer on the boundary between The Inner Circle and Rising Stars genuinely shares characteristics of both segments. This structural continuity means that metric scores moderate by standard benchmarks can still correspond to highly actionable, business-valid segments. The primary validation criterion is not metric optimality in isolation, but the combination of metric acceptability and demonstrably different optimal marketing strategies for each segment — a standard that the K=6 solution satisfies.

8. Results: The Six Customer Segments

The K=6 GMM identifies six behaviourally distinct customer archetypes. Each segment is profiled below with its quantitative characteristics, naming rationale, and the behavioural logic that underpins the assigned persona. All percentage figures are validated against the client's presentation materials.

8.1 Revenue and Size Overview

Segment	% Customers	~N	% Revenue	Est. Revenue	Defining KPI
■ The Inner Circle	25.6%	5,498	42.8%	~€5.12M	Highest monetary + frequency; lowest avg_discount_pct
■ Rising Stars	28.3%	6,078	20.2%	~€2.41M	Stable mid-tier behaviour; high CLV upgrade potential
■■ Style Explorers	15.6%	3,350	13.7%	~€1.64M	High n_categories_explored; cross-category breadth
■■ Deal Chasers	14.2%	3,050	11.9%	~€1.42M	74.6% promotional transaction rate
■ Dormant Potential	11.1%	2,385	3.8%	~€0.45M	Highest recency_days; lowest engagement
■ Quality Seekers	5.2%	1,113	7.6%	~€0.91M	33% return rate; above-avg basket value
TOTAL	100%	21,477	100%	€11.96M	

Table 8: Segment size, revenue contribution, and primary behavioural differentiator.

8.2 ■ The Inner Circle — 25.6% of Customers, 42.8% of Revenue

The Inner Circle is the retailer's highest-value customer cohort: approximately 5,498 individuals who collectively generate 42.8% of observed revenue (~€5.12M). Their defining profile combines high purchase frequency, high average order value, minimal recency (they purchased recently), and critically, the lowest avg_discount_pct of any segment — indicating a willingness to pay full price that materially exceeds the retailer's average. This full-price purchasing behaviour is the single most commercially significant characteristic: revenue from this segment carries the highest net margin per euro generated, entirely unlike Deal Chasers whose nominally similar basket values are heavily discounted.

Naming rationale: The name 'Inner Circle' reflects the exclusive status of this group and the retailer's existential financial dependency on it. The extreme revenue concentration — 25.6% of customers generating 42.8% of revenue, more concentrated than the standard Pareto 80/20 rule — means that a 5 percentage-point churn increase in this segment alone would eliminate approximately €256K in annual revenue, exceeding the entire proposed strategic investment of €140K/year. This concentration risk transforms what would ordinarily be a retention strategy into a financial risk management imperative.

8.3 ■ Rising Stars — 28.3% of Customers, 20.2% of Revenue

Rising Stars is simultaneously the largest segment by customer count (~6,078) and the second largest by revenue (~€2.41M). Their behavioural profile is characterised by stable, moderate-frequency purchasing at moderate average order values, without strong promotional dependency. This combination — regular purchasing, modest basket, low discount dependency — is the hallmark of a customer whose relationship with the brand is developing but not yet at VIP maturity.

Naming rationale: The strategic significance of Rising Stars lies in their position as the natural feeder pool for The Inner Circle. The CLV model estimates that a Rising Stars customer successfully upgraded to Inner Circle behaviour generates approximately €534 in incremental CLV. Given the segment's size (~6,078 customers), even a 5% conversion rate represents 304 customers $\times$ €534 = ~€162K in CLV uplift, at an estimated campaign cost of €18K/year. 'Rising Stars' was chosen over alternative names (e.g., 'Regular Buyers') precisely because it encodes the upgrade trajectory as the primary strategic lens for this group.

8.4 ■■ Style Explorers — 15.6% of Customers, 13.7% of Revenue

Style Explorers are defined by high category breadth — they purchase across multiple product categories — and exhibit cross-category exploration behaviour. Their monetary value is moderate, but their category affinity profile reveals a qualitatively different kind of brand engagement: they are not anchored to a single product type but are actively exploring the brand's product range. A customer already purchasing across Tops, Bottoms, and Accessories has implicitly signalled affinity with the brand's overall aesthetic, not merely a single product's utility.

Naming rationale: The name reflects their defining behaviour pattern. The strategic implication is that this segment is the most receptive to editorial marketing content (lookbook campaigns, outfit-completion recommendations, seasonal style guides) that would be ineffective on, for example, Dormant Potential customers who have not yet established multi-category purchase behaviour.

8.5 ■■ Deal Chasers — 14.2% of Customers, 11.9% of Revenue

Deal Chasers present the most structurally complex challenge in the segmentation. They purchase at high average basket values (€249 per transaction) but 74.6% of their transactions involve a promotional discount. The paradox is that they appear commercially valuable by gross revenue but their net margin contribution is substantially lower than the headline figures imply. More critically, systematic discounting to this segment creates promotional conditioning: customers learn through repeated experience to defer purchasing until a discount is available, progressively increasing the minimum discount threshold required to trigger a purchase. Left unaddressed, this dynamic erodes margin on the retailer's highest-ticket items indefinitely.

Naming rationale: 'Deal Chasers' is deliberately neutral rather than pejorative. These customers are not intrinsically low-value — their average basket of €249 is above the dataset average, and they spend significantly when promotions are available. The challenge is re-conditioning their purchasing behaviour toward full-price engagement through loyalty incentives, exclusivity framing, and a gradual promotional step-down strategy. The name signals the diagnosis (discount dependency) without presupposing the customer's intention.

8.6 ■ Dormant Potential — 11.1% of Customers, 3.8% of Revenue

Dormant Potential customers exhibit the highest `recency_days` of any segment (they have not purchased recently), the lowest purchase frequency, and minimal monetary contribution. This segment is the consolidated result of four GMM components that shared identical low-engagement profiles — high inactivity, low frequency, low spend. Despite their low revenue contribution (3.8%), their presence in the CRM database confirms a historical purchase relationship with the brand, which has three important implications: (i) they are known to the retailer's systems; (ii) reactivation cost is lower than cold acquisition; (iii) their prior purchase history provides a preference signal for targeted reactivation messaging.

Naming rationale: The compound name encodes both the diagnostic reality ('Dormant') and the commercial opportunity ('Potential'). This framing is intentional: it prevents the segment from being dismissed as a lost cause while accurately representing the reactivation challenge. A 15% reactivation rate on 2,385 customers at €189 average first-purchase spend generates approximately €68K in incremental revenue from a win-back campaign costing ~€10K/year — a 6.8× gross return multiple and the single highest ROI initiative in the portfolio.

8.7 ■ Quality Seekers — 5.2% of Customers, 7.6% of Revenue

Quality Seekers are the smallest segment by customer count (5.2%, ~1,113 customers) but their revenue contribution per customer is disproportionately high: 7.6% of revenue from 5.2% of customers implies an above-average spend rate. Their defining characteristic is a 33% return rate — equivalent to 1 in 3 items returned, the highest of any segment — combined with above-average basket values on non-returned items. This behavioural pattern reflects a well-documented phenomenon in fashion retail: customers with high quality standards and specific fit requirements who engage in a trial-and-selection purchasing process, ordering multiple sizes or styles to identify the optimal item.

Naming rationale: The name 'Quality Seekers' encodes the hypothesis that their return behaviour is driven by high standards and genuine fit uncertainty rather than opportunistic purchasing with intent to return. This hypothesis is supported by their above-average spend on retained items: a customer returning 33% of purchases by volume but generating above-average revenue from the 67% retained is a genuinely engaged buyer, not a serial returner. The correct intervention is therefore informational — specifically a data-driven Size Advisor Engine that leverages this customer's own purchase-and-return history to surface personalised size recommendations — rather than punitive (return fees or restrictions), which would risk alienating a high-revenue cohort.

9. Business Implications and Strategic Recommendations

9.1 Revenue Concentration Risk

The most structurally significant finding of the segmentation is the extreme revenue concentration in The Inner Circle: 25.6% of customers generate 42.8% of total revenue. This concentration is materially more severe than the canonical Pareto 80/20 rule and has direct implications for how the retailer should prioritise its marketing investment. A 5 percentage-point increase in Inner Circle churn rate would reduce annual revenue by approximately €256K. This single figure — €256K in potential loss from a single 5pp churn shock to 25.6% of the customer base — motivates treating Inner Circle retention not as a marketing option but as a financial risk management priority, warranting the highest per-customer investment of any segment.

9.2 Per-Segment Strategy Matrix

Segment	Primary Strategy	Key Initiative	Budget (est.)
■ The Inner Circle	Retention + Exclusivity	VIP loyalty programme with exclusive access	€45K/yr
■ Rising Stars	Upgrade Path Activation	Gamified loyalty programme: personalised 'You're 30% closer to Silver tier' milestone email	€30K/yr
■ Style Explorers	Cross-Sell & Bundle	'Complete the Look' recommendation engine	€20K/yr
■ Deal Chasers	Margin Protection	A/B test on two equal sub-groups: Control (standard 20% discount email); Treatment	€20K/yr
■ Dormant Potential	Reactivation	Three-touch win-back sequence: Day 30 — 'Hi [Name], your [purchased category]? + con	€10K/yr
■ Quality Seekers	Fit & Information Tools	ML-powered Size Advisor Engine on Product Detail Pages: personalised size recomm	€10K/yr
TOTAL			€140K/yr

Table 9: Per-segment strategy matrix with directional initiative cost estimates.

9.3 Customer Lifetime Value Projection

Discounted CLV was computed per customer using the formula in §2.3. Segment-level CLV aggregates to a total projected portfolio value of **€15.3M**, representing a 27.9% premium over the €11.96M observed revenue — reflecting the forward-looking value of successfully retaining and upgrading the customer base. The Inner Circle accounts for the largest share of projected CLV, further confirming the disproportionate return on investment from retention initiatives targeting that segment. CLV projections by segment are visualised in assets/clv_analysis.png.

9.4 Revenue at Risk: Scenario Analysis

A symmetric $\pm 20\%$ perturbation of churn rates was applied to each segment to stress-test the revenue model under adverse and favourable conditions. The $\pm 20\%$ boundary represents a reasonable upper bound for marketing-driven churn variation within a 12-month horizon. Key findings: (i) The Inner Circle dominates the downside risk profile — a 20% churn rate increase generates the largest absolute revenue reduction of any segment. (ii) Rising Stars presents the largest combined risk-and-opportunity profile given its size. (iii) For Dormant Potential, the churn perturbation framework is less meaningful; the relevant metric is reactivation rate, modelled separately.

9.5 ROI Framework: €140K → +€718K → 5.1×

The aggregate €140K/year strategic investment across all six initiatives was validated against projected incremental revenue uplift. Uplift estimates are derived from segment-specific conversion and retention improvement assumptions applied to the addressable customer population in each segment. The resulting projected uplift of **+€718K/year** implies a **5.1× gross return multiple**. This figure should be interpreted as a directional decision-support estimate, not an audited financial forecast. It is presented to demonstrate that the proposed reallocation of marketing spend from undifferentiated mass campaigns toward segment-specific initiatives is economically justified under conservative assumptions about campaign execution quality and customer response rates. Full ROI breakdown is visualised in `assets/roi_framework.png`.

10. Deployment Considerations

The CRISP-DM framework's sixth phase addresses the transition from a validated analytical model to a production system generating business value on an ongoing basis. While a full cloud deployment is beyond the scope of this academic project, the architectural recommendations below reflect sound MLOps practice for a CRM integration context.

10.1 Scoring New Customers

The segmentation pipeline comprises three serialised objects: the RobustScaler, the UMAP reducer, and the GMM classifier, each persisted as a pickle file in the `models/` directory. To assign a new customer to a segment, the inference sequence is:

```
new_features → scaler.transform() → reducer.transform() → gmm.predict()
```

This pipeline can be exposed as a microservice REST API that receives a customer feature vector (computed from transactional data upstream) and returns a segment assignment and a probability vector across all six components. The probabilistic output is a key advantage of GMM over hard assignment methods: customers with low maximum probability score are genuine behavioural boundary cases, and may warrant deferred classification or a blended treatment strategy until their behavioural profile stabilises.

10.2 Re-Training Cadence

The model should be re-trained when any of the following conditions are met:

- Cluster size distribution drifts by more than 10% from the baseline over a rolling 90-day window (e.g., The Inner Circle declining from 25.6% to below 23%).
- A scheduled quarterly re-training cycle is completed, incorporating the most recent 24-month rolling data window to reflect evolved purchasing patterns.
- The Silhouette Score on the current production dataset falls below 80% of the training-time score, indicating that the UMAP embedding no longer accurately represents the current behavioural manifold.

10.3 CRM Integration

The model output — `customer_segmentation_results.csv` with fields `cluster_gmm`, `cluster_name`, `cluster_confidence`, and all 32 behavioural KPIs — is designed for direct ingestion into a CRM platform (e.g., Salesforce, Adobe Campaign, or Jakala's proprietary Loyalty/Activation stack). The `cluster_name` field maps directly to campaign audience definitions. The `cluster_confidence` field enables high-certainty-first campaign rollout: targeting the top 50% of each segment by confidence in a pilot phase before scaling to the full segment reduces the risk of early-stage campaign errors affecting borderline customers.

11. Limitations

- **Single-retailer scope:** The segmentation was developed and validated on data from a single Italian fashion retailer. The specific segment profiles, KPI thresholds, and persona characteristics reflect this retailer's particular customer base and product mix. The methodology is transferable; the segment definitions are not, and should not be applied directly to other retailers without re-training.
- **24-month observation window:** Customers active before March 2023 but dormant within the window may be misclassified as Dormant Potential when they may be seasonal buyers with multi-year cycles. A longer observation window would improve the recency signal for such customers.
- **Demographic features excluded from clustering:** Age, gender, and geography were excluded from the clustering input to ensure that segments are driven by behavioural patterns, not demographics. While this is the methodologically defensible choice for a behavioural segmentation, it means the model does not capture demographic heterogeneity within segments — an important caveat for campaigns where channel selection depends on age or geography.
- **UMAP non-determinism:** Despite fixing `random_state=42`, UMAP's spectral initialisation introduces minor numerical non-determinism across different hardware environments and operating system versions. Results may differ marginally when the pipeline is re-run on different machines.
- **Production deployment not implemented:** The deployment section describes an architectural approach rather than a working production system. Full deployment requires cloud infrastructure, API development, CRM integration, data governance workflows, and GDPR-compliant data handling procedures beyond the scope of this academic project.
- **Financial projections are directional:** CLV, Revenue at Risk, and ROI estimates are based on segment-level assumptions that have not been validated against historical campaign lift data. They are presented as structured decision-support tools, not audited financial forecasts.

12. Conclusions

This project demonstrates that unsupervised machine learning — specifically a pipeline of RobustScaler normalisation, UMAP dimensionality reduction, and Gaussian Mixture Model clustering — can transform a heterogeneous transactional database into a structured, actionable segmentation framework for a fashion retailer. The three central conclusions are as follows.

- **The customer base is structurally concentrated.** 25.6% of customers generate 42.8% of revenue — a concentration more severe than the standard Pareto rule. This makes Inner Circle retention a financial risk management priority rather than merely a marketing objective. Any undifferentiated approach that fails to distinguish and protect this cohort creates disproportionate revenue exposure.
- **Behavioural segments require qualitatively different marketing levers.** The six segments are not simply high/medium/low value tiers. Deal Chasers and Quality Seekers both generate above-average transaction values but require diametrically opposite interventions: margin protection vs. informational fit tools. A coarser segmentation that collapses these into a single category would prescribe contradictory strategies, rendering the segmentation operationally useless.
- **Data-driven segmentation is financially justified.** The proposed €140K/year investment in segment-specific initiatives is projected to generate +€718K/year in incremental revenue, a 5.1× gross return multiple. Even under conservative assumptions, the directional argument for moving from mass marketing to targeted, data-driven engagement is robust.

From a broader perspective, this project illustrates the value creation mechanism of data as a strategic intangible asset in the digital economy (Corrado et al., 2009; Haskel and Westlake, 2018). The same 24 months of transactional records that existed in the retailer's CRM database without analytical structure can, through the application of AI-driven segmentation methods, yield a concrete and measurable marketing capability. The data itself has not changed; its **economic value has been unlocked through analytical investment** — a direct illustration of how digitalisation converts raw information into a productive intangible asset on the firm's balance sheet. The resulting capability — a segmented, CLV-weighted customer map with segment-specific intervention playbooks — constitutes proprietary intelligence that competitors without equivalent analytical infrastructure cannot replicate. This is the core economic logic of AI adoption in digital commerce: it raises **marketing productivity** (more revenue per euro of marketing spend) and improves **allocative efficiency** (budget directed to where the marginal return is highest, not distributed uniformly). The €140K/year investment generating a 5.1× gross return multiple is the quantified expression of that productivity gain.

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